

AMATH 482/582: HOMEWORK 1

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ABSTRACT. This project uses Fourier analysis and the Fast Fourier Transform to locate and track a moving submarine using noisy three-dimensional acoustic data. By identifying the dominant spatial frequency of the submarine signal, a narrow-band Fourier filter was applied to reduce noise. Transforming the filtered data back to physical space produced a clear, localized signal at each time step, allowing the submarine's position to be tracked over 24 hours. The resulting trajectory is smooth and continuous, showing that frequency-domain filtering is an effective way to remove noise and recover meaningful structure from the data.

1. INTRODUCTION AND OVERVIEW

In this homework, Fourier analysis and the Fast Fourier Transform (FFT) are used to analyze noisy three-dimensional acoustic data and track the motion of a submarine. The data consist of spatial pressure measurements collected over time, and the goal is to identify the submarine's signal, filter out noise, and reconstruct its path over a 24-hour period.

The main idea is to work in the frequency domain, where the submarine signal can be separated from background noise. First, the dominant spatial frequency associated with the submarine is identified. This frequency is then used to build a narrow-band Fourier filter that removes most of the noise while keeping the submarine signal. After applying the inverse Fourier transform, the submarine appears as a localized peak in space at each time step, making it possible to track its position over time. This approach shows how Fourier-based methods can be used to clean noisy data and extract useful physical information.

2. THEORETICAL BACKGROUND

Task 1: Frequency Signature

The submarine acoustic pressure data is pretty jumbled at first. It contains both the actual signal and a lot of noise, and since the data is four-dimensional, it is difficult to work with in its raw form. Because of this, it is hard to identify any dominant behavior directly, and the noise makes simple averaging ineffective.

For the time being, this is a snapshot of the data represented as a function.

$$f(x, y, z) : [0, 2\pi)^3 \rightarrow \mathbb{R},$$

To make the data usable, we first apply a Fourier transform. This reshapes the data into an oscillatory form, which makes it much easier to separate meaningful frequency information from noise. In the frequency domain, the signal emitted by the submarine appears as consistent frequency components, while the noise is spread out and less structured.

For a fixed time instance, the acoustic pressure field can be represented using a three-dimensional Fourier series

$$f(x, y, z) = \sum_{k \in \mathbb{Z}^3} \hat{f}(k) e^{i\langle k, x \rangle},$$

where $k = (k_x, k_y, k_z)$ is the frequency vector and $\langle k, x \rangle = k_x x + k_y y + k_z z$. The Fourier coefficients are defined as

$$\hat{f}(k) = \frac{1}{(2\pi)^3} \int_{[0, 2\pi]^3} f(x) e^{-i\langle k, x \rangle} dx.$$

In practice, these coefficients are computed numerically using the multidimensional Fast Fourier Transform (FFT), which applies one-dimensional FFTs sequentially along each spatial direction.

A fundamental property of the Fourier transform is that adding mean-zero white noise to a signal results in mean-zero white noise in the Fourier coefficients, according to the homework hint.

Let $\hat{f}_j(k)$ denote the Fourier transform of the data at time t_j . The averaged spectrum is defined as

$$\bar{f}(k) = \frac{1}{N} \sum_{j=1}^N \hat{f}_j(k),$$

where N is the number of time snapshots. Averaging becomes useful in the Fourier transform because the submarine's signal stays at the same frequencies over time, so averaging reinforces these components. The noise, on the other hand, oscillates randomly and averages out toward zero. As a result, the average spectrum reveals the center frequency of the submarine while reducing the effect of noise. This is shown below:

By looking at where the averaged power spectrum reaches its maximum, we can identify this dominant frequency.

$$P(k) = |\bar{f}(k)|^2$$

therefore identifies the submarine's center frequency. The magnitude of the frequency vector,

$$|k_0| = \sqrt{k_{0x}^2 + k_{0y}^2 + k_{0z}^2},$$

quantifies the dominant oscillation strength of the acoustic signal.

Task 2: Filtering

Theoretical Background (Task 2). In Task 1, the dominant spatial frequency k^* associated with the submarine signal was identified using the Fourier transform. In Task 2, this frequency information is used to recover the submarine's position as a function of time.

Fourier-domain filtering. For each time snapshot t_j , let

$$f_j(x), \quad x \in [0, 2\pi]^3,$$

denote the measured signal with Fourier coefficients $c_k^{(j)}$. To isolate the submarine contribution, define a frequency filter supported near the dominant mode k^* ,

$$\chi(k) = \begin{cases} 1, & |k - k^*| \leq \delta, \\ 0, & \text{otherwise,} \end{cases}$$

and apply it to the Fourier coefficients:

$$\tilde{c}_k^{(j)} = \chi(k) c_k^{(j)}.$$

Inverse Fourier transform and localization. By linearity of the Fourier transform, applying the inverse Fourier transform to the filtered coefficients yields

$$\tilde{f}(x, t_j) = \sum_{k \in \mathbb{Z}^3} \tilde{c}_k^{(j)} e^{i\langle k, x \rangle}.$$

Since only frequencies near k^* are retained, the resulting signal $\tilde{f}(x, t_j)$ is spatially localized and dominated by the submarine.

Position extraction. The submarine position at time t_j is estimated by the spatial location where the magnitude of the filtered signal is maximal:

$$(x(t_j), y(t_j), z(t_j)) = \arg \max_x |\tilde{f}(x, t_j)|.$$

Trajectory reconstruction. Repeating this procedure for all time snapshots produces the coordinate functions

$$x(t), \quad y(t), \quad z(t),$$

which together define the reconstructed submarine trajectory in space.

Task 3: x, y Coordinates

Now, I use the same reconstructed trajectory used in task 1 and 2 to create the motion on the x - y plane.

Plotting $x(t)$ and $y(t)$ as functions of time provides information about the velocity and directional changes of the submarine. Additionally, plotting y versus x gives a planar projection of the trajectory, allowing visualization of the submarine's horizontal path.

3. ALGORITHM IMPLEMENTATION AND DEVELOPMENT

All computations in this project were performed using Google Colab, which provides a Python-based environment suitable for numerical computing and visualization. The implementation relied on standard scientific computing libraries, including NumPy for array operations and Fast Fourier Transforms. Three-dimensional FFTs were computed using NumPy's built-in FFT routines, following the methods discussed in lecture.

The algorithm was implemented by applying the FFT to each time snapshot of the data, identifying the dominant frequency, and constructing a narrow-band frequency-domain filter. After filtering, the inverse FFT was used to transform the data back to physical space, where the submarine's position was extracted as the location of maximum signal amplitude. This process was repeated for all time steps to reconstruct the submarine's trajectory. All figures shown in this report were generated directly from the Colab notebook, and no code is included here as instructed.

4. COMPUTATIONAL RESULTS

Task 1: Frequency Signature

Previously, I applied the Fourier transform to the data. Here, I average each time snapshot, which reduces the noise and reveals the dominant frequency component, as discussed above. Apply `fftshift` so that the zero frequency component is centered in Fourier space. When looking at a flat cross-section of the averaged frequency data, you can clearly see one frequency that stays in the same place over time.

For each time snapshot, the power spectrum was computed as

$$P_j(k) = |\hat{f}_j(k)|^2,$$

I can clearly see one frequency that stays in the same location over time. This shows that the strongest repeating sound in the data occurs at this specific combination of spatial frequencies.

$$k_0 \approx (-2.20, -5.03, 6.91).$$

The corresponding magnitude of the center frequency is

$$|k_0| \approx 8.82.$$

The magnitude represents the combined effect of the specific frequency components identified above and shows how strong the oscillation is based on its distance from zero frequency. Since this dominant frequency is far from zero, it indicates that the submarine is actively emitting a consistent sound. I identified the strongest repeating signal in the data, confirming that it remains stable over time, and showed that it comes from an actively oscillating source, which can now be used to filter the data and track the submarine.

Task 2: Filtering

The original data was messy, but after applying the narrow-band Fourier filter, it kept the frequencies associated with the submarine and removed most of the noise. When transforming the filtered data back to physical space, the submarine appeared as a clear, localized peak at each time step. The 3D plot shows the submarine's reconstructed path through space. The path is smooth and continuous, which shows that the filtering successfully removed random noise. There are no sudden jumps, meaning the detection is stable and reliable. This confirms that the dominant Fourier mode captures the submarine signal rather than the noise, as shown in figure 1.

Submarine 3D path (from filtered data)

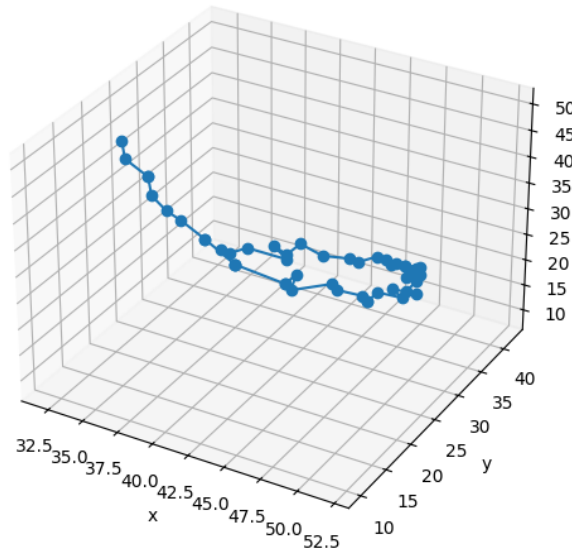


FIGURE 1. Task 2

Task 3: x, y Coordinates

Figure 2 shows the reconstructed x -coordinate as a function of time. The plot exhibits an approximately linear increase over the 24-hour period, indicating nearly constant velocity in the x -direction.

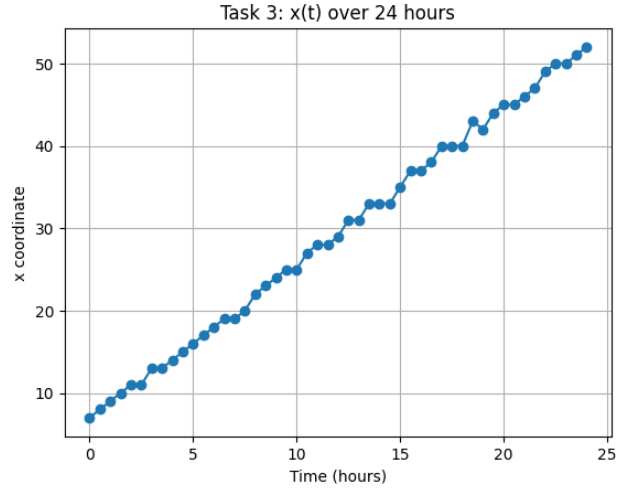
FIGURE 2. $x(t)$ over 24 hours

Figure 3 shows the reconstructed y -coordinate as a function of time. Unlike $x(t)$, the y -coordinate displays non-monotonic behavior. Initially, $y(t)$ remains relatively constant, followed by a steady decrease to a minimum value, and finally a slight increase near the end of the time interval. This behavior indicates a gradual change in the submarine's direction in the y -direction rather than the same motion.

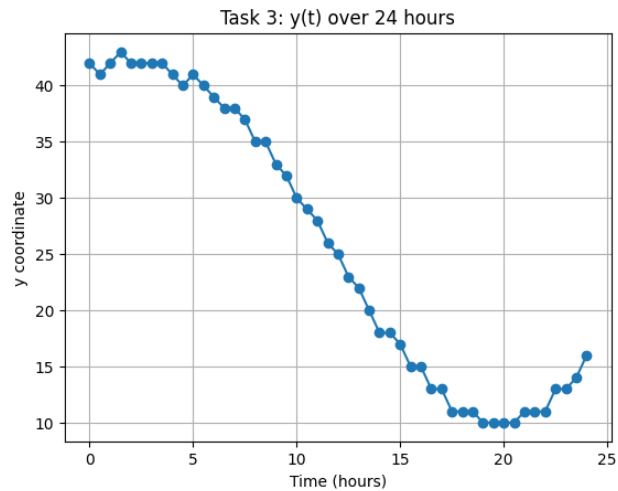
FIGURE 3. $y(t)$ over 24 hours

Figure 4 shows the submarine trajectory projected onto the x - y plane. The curve is smooth and continuous, confirming that the Fourier filtering successfully removed noise while preserving the motion. The shape of the path reflects the combined effects of steady motion in x and varying motion in y , resulting in a curved horizontal trajectory.

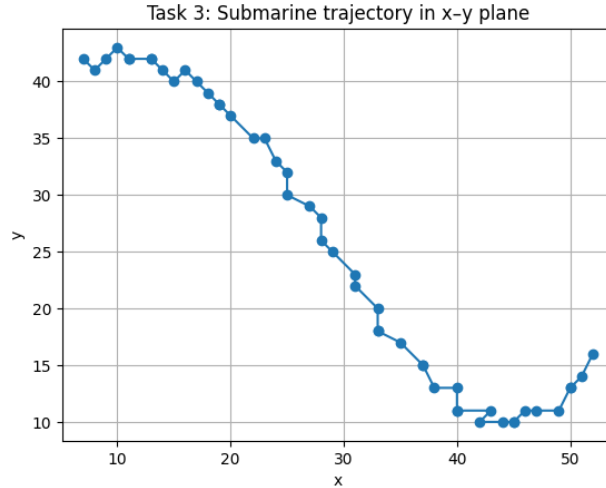


FIGURE 4. Submarine trajectory in x-y plane

Together, these plots demonstrate that the reconstructed submarine motion is physically consistent and that Fourier-based filtering and reconstruction worked for tracking the submarine's position over time.

5. SUMMARY AND CONCLUSIONS

The data given initially was difficult to read, so I used the Fourier transform to analyze the noisy 3D acoustic data and track the motion of the submarine. The initial data was messy because the noise was combined with the submarine's signal. By transforming the data into the frequency domain, the dominant spatial frequency associated with the submarine was identified after averaging, which allowed it to be separated from the background noise. Using this cleaner version of the data, a narrow-band Fourier filter was applied to each time snapshot, and the inverse Fourier transform produced a localized signal.

The submarine's position was found at the location of maximum amplitude, making it possible to reconstruct the full 3D trajectory over time. The results show that the path is smooth and continuous, meaning that the filter successfully removed noise while preserving the submarine signal. The submarine moves steadily in the x-direction, while the y-coordinate shows non-monotonic behavior, suggesting a gradual change in direction rather than straight-line motion. The depth coordinate varies smoothly with no abrupt jumps, further confirming the stability of the detection. Overall, these results demonstrate that isolating the dominant Fourier mode and applying frequency-domain filtering provides an effective and reliable approach for tracking coherent structures in noisy data.

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